

Linear Algebra: Comprehensive Problem Solutions

Complete Problem Sets 1–5 (Vectors, Matrices, Linear Spaces, Operators, and SVD/Tensors)

Linear Algebra Coursework

2024

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Part I

Assignment 1. Vectors and Linear Systems

1 Section 1. Linear Combinations and Vector Equations

Problem 1.1

Problem Statement: Find the linear combination $7a + 3b - 4c$ for the following vectors:

$$a = (3, -1, 3), \quad b = (5, 2, 7), \quad c = (-3, -2, 6)$$

Solution: Multiply each vector by its corresponding scalar and perform component-wise addition:

$$\begin{aligned} 7a &= 7 \cdot (3, -1, 3) = (21, -7, 21), \\ 3b &= 3 \cdot (5, 2, 7) = (15, 6, 21), \\ -4c &= -4 \cdot (-3, -2, 6) = (12, 8, -24). \end{aligned}$$

Summing the results:

$$\begin{aligned} 7a + 3b - 4c &= (21 + 15 + 12, -7 + 6 + 8, 21 + 21 - 24) \\ &= (48, 7, 18). \end{aligned}$$

Answer: $(48, 7, 18)$.

—

Problem 1.2

Problem Statement: Solve for the vector x in the given equations:

a) $a + 9b + x - c = 0$, where $a = (3, 5, 1, 6)$, $b = (2, 9, 4, 2)$, $c = (3, 2, 8, 4)$.

Solution: Isolate vector x :

$$x = c - a - 9b.$$

Calculate $9b$:

$$9b = 9 \cdot (2, 9, 4, 2) = (18, 81, 36, 18).$$

Compute the components of x :

$$\begin{aligned} x_1 &= 3 - 3 - 18 = -18, \\ x_2 &= 5 - 2 - 81 = -84, \\ x_3 &= 1 - 8 - 36 = -29, \\ x_4 &= 6 - 4 - 18 = -20. \end{aligned}$$

Answer: $x = (-18, -84, -29, -20)$.

b) $3.5(a - x) - 2(b + x) + 1.5(c + 3x) = 0$, where $a = (3, 5, 4, 2)$, $b = (2, 3, 4, 2)$, $c = (5, 2, 9, 3)$.

Solution: Expand and group the terms containing x :

$$\begin{aligned} 3.5a - 3.5x - 2b - 2x + 1.5c + 4.5x &= 0 \\ (-3.5 - 2 + 4.5)x + 3.5a - 2b + 1.5c &= 0 \\ -x + 3.5a - 2b + 1.5c &= 0 \implies x = 3.5a - 2b + 1.5c. \end{aligned}$$

Compute the components of x :

$$x_1 = 3.5(3) - 2(2) + 1.5(5) = 10.5 - 4 + 7.5 = 14,$$

$$x_2 = 3.5(5) - 2(3) + 1.5(2) = 17.5 - 6 + 3 = 14.5,$$

$$x_3 = 3.5(4) - 2(4) + 1.5(9) = 14 - 8 + 13.5 = 19.5,$$

$$x_4 = 3.5(2) - 2(2) + 1.5(3) = 7 - 4 + 4.5 = 7.5.$$

Answer: $x = (14, 14.5, 19.5, 7.5)$.

—

Problem 1.3

Problem Statement: Given vectors $a = (1, 2)$ and $b = (4, 3)$. Their linear combination yields $c = (57, 45)$. Find the coefficients.

Solution: Set up the linear combination equation $\alpha a + \beta b = c$:

$$\begin{cases} \alpha + 4\beta = 57 \\ 2\alpha + 3\beta = 45 \end{cases}$$

Multiply the first equation by 2:

$$2\alpha + 8\beta = 114$$

Subtract the second equation:

$$(2\alpha + 8\beta) - (2\alpha + 3\beta) = 114 - 45 \implies 5\beta = 69 \implies \beta = \frac{69}{5} = 13.8.$$

Find α :

$$\alpha = 57 - 4(13.8) = 57 - 55.2 = 1.8 = \frac{9}{5}.$$

Answer: $\alpha = 1.8$ (or $\frac{9}{5}$), $\beta = 13.8$ (or $\frac{69}{5}$).

—

Problem 1.4

Problem Statement: Find α such that vectors $a = (3, \alpha, 2)$ and $b = (2 - \alpha, -1, 5)$ are orthogonal.

Solution: Vectors are orthogonal if and only if their dot product is zero:

$$\langle a, b \rangle = 0$$

$$3(2 - \alpha) + \alpha(-1) + 2(5) = 0$$

$$6 - 3\alpha - \alpha + 10 = 0$$

$$16 - 4\alpha = 0 \implies \alpha = 4.$$

Answer: $\alpha = 4$.

2 Section 2. Norms, Angles, and Vector Properties

Problem 2.1

Problem Statement: Compute the Manhattan (ℓ_1) and Euclidean (ℓ_2) norms for:

a) $a = (3, 9, 2, 3, 4, 0, -1, -11, 3)$

b) $b = (12, 13, 23, -22, -34, 31, 21)$

Solution:

a)

• **Manhattan Norm (ℓ_1):**

$$\|a\|_1 = |3| + |9| + |2| + |3| + |4| + |0| + |-1| + |-11| + |3| = 36.$$

• **Euclidean Norm (ℓ_2):**

$$\|a\|_2 = \sqrt{3^2 + 9^2 + 2^2 + 3^2 + 4^2 + 0^2 + (-1)^2 + (-11)^2 + 3^2} = \sqrt{250} = 5\sqrt{10} \approx 15.811.$$

b)

• **Manhattan Norm (ℓ_1):**

$$\|b\|_1 = |12| + |13| + |23| + |-22| + |-34| + |31| + |21| = 156.$$

• **Euclidean Norm (ℓ_2):**

$$\|b\|_2 = \sqrt{12^2 + 13^2 + 23^2 + (-22)^2 + (-34)^2 + 31^2 + 21^2} = \sqrt{3884} \approx 62.322.$$

—

Problem 2.2

Problem Statement: Given vectors $a = (1, 2, 3)$ and $b = (4, 3, 5)$, find the angle between them.

Solution:

$$\cos \theta = \frac{\langle a, b \rangle}{\|a\|_2 \|b\|_2}$$

1. Dot product: $\langle a, b \rangle = 1 \cdot 4 + 2 \cdot 3 + 3 \cdot 5 = 25$. 2. Norms: $\|a\|_2 = \sqrt{14}$, $\|b\|_2 = \sqrt{50} = 5\sqrt{2}$. 3. Angle calculation:

$$\cos \theta = \frac{25}{\sqrt{14} \cdot 5\sqrt{2}} = \frac{5}{2\sqrt{7}} \approx 0.94491 \implies \theta = \arccos\left(\frac{5}{2\sqrt{7}}\right) \approx 19.11^\circ.$$

Answer: $\theta \approx 19.11^\circ$.

—

Problem 2.3

Problem Statement: Vector x of length 100 contains elements $x_i \geq 0$ (number of absences for student i). Using the ℓ_1 norm, write formulas for:

a. Total number of absences;

b. Average number of absences.

Solution: Since $x_i \geq 0$, $\|x\|_1 = \sum_{i=1}^{100} |x_i| = \sum_{i=1}^{100} x_i$.

a. **Total Absences:** $S = \|x\|_1$.

b. **Average Absences:** $\bar{x} = \frac{\|x\|_1}{100}$.

—

Problem 2.4

Problem Statement: Prove the identities for any vectors $a, b \in \mathbb{R}^n$:

a) $(a + b)^T(a - b) = \|a\|^2 - \|b\|^2$

Proof:

$$(a + b)^T(a - b) = (a^T + b^T)(a - b) = a^T a - a^T b + b^T a - b^T b.$$

Since $a^T b = b^T a$ (scalar result), the middle terms cancel out:

$$a^T a - b^T b = \|a\|^2 - \|b\|^2. \quad \square$$

b) $\|a + b\|^2 + \|a - b\|^2 = 2(\|a\|^2 + \|b\|^2)$

Proof:

$$\|a + b\|^2 = (a + b)^T(a + b) = \|a\|^2 + 2a^T b + \|b\|^2,$$

$$\|a - b\|^2 = (a - b)^T(a - b) = \|a\|^2 - 2a^T b + \|b\|^2.$$

Summing both equations yields $2\|a\|^2 + 2\|b\|^2 = 2(\|a\|^2 + \|b\|^2)$. $\square$

3 Section 3. Systems of Linear Equations (SLE) and Gaussian Elimination

Problem 3.1

Problem Statement: The perimeter of a triangle is 30 cm. The longest side is 4 cm longer than the medium side. Three times the shortest side equals the sum of the medium and longest sides plus 2 cm. Find the sides using Gaussian elimination.

Solution: Let x be the shortest, y the medium, and z the longest side.

$$\begin{cases} x + y + z = 30 \\ -y + z = 4 \\ 3x - y - z = 2 \end{cases}$$

Augmented matrix reduction:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 30 \\ 0 & -1 & 1 & 4 \\ 3 & -1 & -1 & 2 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - 3R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 30 \\ 0 & -1 & 1 & 4 \\ 0 & -4 & -4 & -88 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - 4R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 30 \\ 0 & -1 & 1 & 4 \\ 0 & 0 & -8 & -104 \end{array} \right)$$

Back-substitution:

- $-8z = -104 \implies z = 13$ cm.
- $-y + 13 = 4 \implies y = 9$ cm.
- $x + 9 + 13 = 30 \implies x = 8$ cm.

Answer: Sides are 8 cm, 9 cm, and 13 cm.

—

Problem 3.2

Problem Statement: Solve the linear systems using Gaussian elimination:

a)

$$\begin{cases} 3x_1 + 2x_2 + x_3 = 0 \\ 7x_1 + 6x_2 + 5x_3 = 0 \\ 5x_1 + 4x_2 + 3x_3 = 0 \end{cases}$$

Solution:

$$\left(\begin{array}{ccc} 3 & 2 & 1 \\ 7 & 6 & 5 \\ 5 & 4 & 3 \end{array} \right) \xrightarrow{\begin{array}{l} R_2 \leftarrow 3R_2 - 7R_1 \\ R_3 \leftarrow 3R_3 - 5R_1 \end{array}} \left(\begin{array}{ccc} 3 & 2 & 1 \\ 0 & 4 & 8 \\ 0 & 2 & 4 \end{array} \right) \xrightarrow{R_3 \leftarrow 2R_3 - R_2} \left(\begin{array}{ccc} 3 & 2 & 1 \\ 0 & 4 & 8 \\ 0 & 0 & 0 \end{array} \right)$$

Set $x_3 = t$ (free variable):

- $4x_2 + 8t = 0 \implies x_2 = -2t$.
- $3x_1 + 2(-2t) + t = 0 \implies x_1 = t$.

Answer: $(x_1, x_2, x_3) = (t, -2t, t)$, for $t \in \mathbb{R}$.

b)

$$\begin{cases} -3x_1 + x_2 + x_3 = 0 \\ 5x_1 + x_2 - 2x_3 = 2 \\ -2x_1 - 2x_2 + x_3 = -3 \end{cases}$$

Solution:

$$\left(\begin{array}{ccc|c} -3 & 1 & 1 & 0 \\ 5 & 1 & -2 & 2 \\ -2 & -2 & 1 & -3 \end{array} \right) \xrightarrow[R_3 \leftarrow -3R_3 - 2R_1]{R_2 \leftarrow -3R_2 + 5R_1} \left(\begin{array}{ccc|c} -3 & 1 & 1 & 0 \\ 0 & 8 & -1 & 6 \\ 0 & -8 & 1 & -9 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|c} -3 & 1 & 1 & 0 \\ 0 & 8 & -1 & 6 \\ 0 & 0 & 0 & -3 \end{array} \right)$$

Row 3 gives $0 = -3$ (inconsistent).

Answer: No solution.

c)

$$\begin{cases} 7x_1 - 5x_2 - 2x_3 - 4x_4 = 8 \\ -3x_1 + 2x_2 + x_3 + 2x_4 = -3 \\ 2x_1 - x_2 - x_3 - 2x_4 = 1 \\ -x_1 + x_3 + 2x_4 = 1 \end{cases}$$

Solution: Reordering rows putting R_4 first:

$$\left(\begin{array}{cccc|c} -1 & 0 & 1 & 2 & 1 \\ 2 & -1 & -1 & -2 & 1 \\ -3 & 2 & 1 & 2 & -3 \\ 7 & -5 & -2 & -4 & 8 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} -1 & 0 & 1 & 2 & 1 \\ 0 & -1 & 1 & 2 & 3 \\ 0 & 2 & -2 & -4 & -6 \\ 0 & -5 & 5 & 10 & 15 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} -1 & 0 & 1 & 2 & 1 \\ 0 & -1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Free variables: $x_3 = s$, $x_4 = t$. Expressing dependent variables: $x_2 = s + 2t - 3$, $x_1 = s + 2t - 1$.

Answer: $(x_1, x_2, x_3, x_4) = (s + 2t - 1, s + 2t - 3, s, t)$, for $s, t \in \mathbb{R}$.

4 Section 4. Bonus Problems and Theory (Assignment 1)

Bonus Problem 1.1

Problem Statement: Given vectors $a = (1, 5, 3)$, $b = (6, -4, -2)$, $c = (0, -5, 7)$, $d = (-20, 27, 35)$. Find scalars α, β, γ such that $\alpha a, \beta b, \gamma c, d$ form a closed polygonal chain.

Solution: Closure requires $\alpha a + \beta b + \gamma c + d = 0 \implies \alpha a + \beta b + \gamma c = -d$:

$$\begin{cases} \alpha + 6\beta = 20 \\ 5\alpha - 4\beta - 5\gamma = -27 \\ 3\alpha - 2\beta + 7\gamma = -35 \end{cases}$$

Solving the system yields $\alpha = -4$, $\beta = 4$, $\gamma = -2$.

Answer: $\alpha = -4$, $\beta = 4$, $\gamma = -2$.

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Bonus Problem 1.2

Problem Statement: Solve $Ax = 12x$, where $A = \begin{pmatrix} 6 & 4 & 3 \\ 6 & 0 & 9 \\ 0 & 8 & 0 \end{pmatrix}$ and $\sum_{i=1}^3 x_i = 1$.

Solution: Rewrite as $(A - 12I)x = 0$:

$$\begin{pmatrix} -6 & 4 & 3 \\ 6 & -12 & 9 \\ 0 & 8 & -12 \end{pmatrix} \rightarrow \begin{pmatrix} -6 & 4 & 3 \\ 0 & -8 & 12 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus $x_2 = 1.5x_3$ and $x_1 = 1.5x_3$. Applying $x_1 + x_2 + x_3 = 1$:

$$1.5x_3 + 1.5x_3 + x_3 = 1 \implies 4x_3 = 1 \implies x_3 = 0.25.$$

Answer: $x = (0.375, 0.375, 0.25)^T$.

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Theoretical Questions

a) When are two vectors orthogonal?

Answer. Two vectors a and b are orthogonal if and only if their inner product equals zero: $\langle a, b \rangle = 0$. □

b) Give an example of data that can be represented as a vector.

Answer. A user profile in a recommendation algorithm (e.g., movie preferences across genres) can be encoded as a feature vector: $x = (\text{Sci-Fi}, \text{Comedy}, \text{Drama}) = (0.9, 0.1, 0.4)$. □

c) If a variable has zero coefficients across all equations in a system, how many solutions can the system have?

Answer. The system can have either ****0 solutions**** (inconsistent) or ****infinitely many solutions****. It can ****never**** have a unique solution because the variable with zero coefficients acts as a free variable that can take any value in the scalar field without affecting any equation. □

Part II

Assignment 2. Matrices and Matrix Operations

5 Section 1. Matrix Algebra

Problem 1.1

Problem Statement: Compute $3A - 2B + C$ for:

$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 5 & 2 \\ 4 & -3 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 0 & -1 \\ -2 & 1 & 4 \end{pmatrix}$$

Solution:

$$3A - 2B + C = \begin{pmatrix} 6 & -3 & 12 \\ 0 & 9 & -6 \end{pmatrix} + \begin{pmatrix} 2 & -10 & -4 \\ -8 & 6 & -2 \end{pmatrix} + \begin{pmatrix} 3 & 0 & -1 \\ -2 & 1 & 4 \end{pmatrix} = \begin{pmatrix} 11 & -13 & 7 \\ -10 & 16 & -4 \end{pmatrix}.$$

Answer: $\begin{pmatrix} 11 & -13 & 7 \\ -10 & 16 & -4 \end{pmatrix}.$
—

Problem 1.2

Problem Statement: Multiply matrices $A \cdot B$:

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ -1 & 3 \\ 0 & -2 \end{pmatrix}$$

Solution:

$$AB = \begin{pmatrix} 1(2) + 3(-1) + (-2)(0) & 1(1) + 3(3) + (-2)(-2) \\ 2(2) + 0(-1) + 4(0) & 2(1) + 0(3) + 4(-2) \end{pmatrix} = \begin{pmatrix} -1 & 14 \\ 4 & -6 \end{pmatrix}.$$

Answer: $AB = \begin{pmatrix} -1 & 14 \\ 4 & -6 \end{pmatrix}.$
—

Problem 1.3

Problem Statement: Find A^T and verify $(A^T)^T = A$ for $A = \begin{pmatrix} 4 & 1 & -3 \\ 2 & 8 & 5 \end{pmatrix}.$

Solution:

$$A^T = \begin{pmatrix} 4 & 2 \\ 1 & 8 \\ -3 & 5 \end{pmatrix} \implies (A^T)^T = \begin{pmatrix} 4 & 1 & -3 \\ 2 & 8 & 5 \end{pmatrix} = A. \quad \square$$

6 Section 2. Determinants and Inverses

Problem 2.1

Problem Statement: Compute the determinant of $A = \begin{pmatrix} 2 & 3 & -1 \\ 4 & 1 & 5 \\ -2 & 0 & 3 \end{pmatrix}$.

Solution: Expand along Row 3:

$$\det(A) = -2 \begin{vmatrix} 3 & -1 \\ 1 & 5 \end{vmatrix} + 3 \begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix} = -2(15 + 1) + 3(2 - 12) = -32 - 30 = -62.$$

Answer: $\det(A) = -62$.

—

Problem 2.2

Problem Statement: Find the inverse A^{-1} for $A = \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix}$.

Solution: $\det(A) = 3(2) - 4(1) = 2 \neq 0$.

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 2 & -4 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ -0.5 & 1.5 \end{pmatrix}.$$

Answer: $A^{-1} = \begin{pmatrix} 1 & -2 \\ -0.5 & 1.5 \end{pmatrix}$.

7 Section 3. Matrix Equations

Problem 3.1

Problem Statement: Solve $AX = B$ for X , where $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$.

Solution: $\det(A) = -2 \implies A^{-1} = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix}$.

$$X = A^{-1}B = \begin{pmatrix} -2 & 1 \\ 1.5 & -0.5 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} -3 & -4 \\ 4 & 5 \end{pmatrix}.$$

Answer: $X = \begin{pmatrix} -3 & -4 \\ 4 & 5 \end{pmatrix}$.

8 Section 4. Bonus Problems and Theory (Assignment 2)

Bonus Problem 2.1

Problem Statement: Prove that $\text{Tr}(AB - BA) = 0$ for square matrices $A, B \in \mathbb{R}^{n \times n}$.

Proof: Using trace linearity and cyclic property $\text{Tr}(AB) = \text{Tr}(BA)$:

$$\text{Tr}(AB - BA) = \text{Tr}(AB) - \text{Tr}(BA) = \text{Tr}(AB) - \text{Tr}(AB) = 0. \quad \square$$

—

Theoretical Questions

a) Does $AB = BA$ hold in general for square matrices?

Answer. No, matrix multiplication is generally non-commutative. Counterexample:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \implies AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

□

b) What happens to a matrix determinant under transposition?

Answer. The determinant remains unchanged: $\det(A^T) = \det(A)$.

□

Part III

Assignment 3. Linear Spaces, Basis, Rank, and Eigenvalues

9 Section 1. Linear Independence and Bases

Problem 1.1

Problem Statement: Test $v_1 = (1, 2, 3)$, $v_2 = (2, 5, 7)$, $v_3 = (3, 7, 10)$ for linear independence.

Solution:

$$\det \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 7 & 10 \end{pmatrix} = 1(50 - 49) - 2(20 - 21) + 3(14 - 15) = 1 + 2 - 3 = 0.$$

Answer: The vectors are linearly dependent ($v_3 = v_1 + v_2$).

—

Problem 1.2

Problem Statement: Expand $x = (5, 9, 14)$ in basis $e_1 = (1, 1, 1)$, $e_2 = (1, 2, 3)$, $e_3 = (2, -1, 1)$.

Solution:

$$\begin{cases} c_1 + c_2 + 2c_3 = 5 \\ c_1 + 2c_2 - c_3 = 9 \\ c_1 + 3c_2 + c_3 = 14 \end{cases} \implies c_1 = 0, c_2 = 4.6, c_3 = 0.2.$$

Answer: Coordinates of x : $(0, 4.6, 0.2)^T$.

10 Section 2. Matrix Rank and Dimension

Problem 2.1

Problem Statement: Find the rank of $A = \begin{pmatrix} 1 & -2 & 3 & 1 \\ 2 & 1 & -1 & 3 \\ 3 & -1 & 2 & 4 \end{pmatrix}$.

Solution:

$$\begin{pmatrix} 1 & -2 & 3 & 1 \\ 2 & 1 & -1 & 3 \\ 3 & -1 & 2 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 5 & -7 & 1 \\ 0 & 5 & -7 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 5 & -7 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Answer: $\text{rank}(A) = 2$.

11 Section 3. Eigenvalues and Eigenvectors

Problem 3.1

Problem Statement: Find eigenvalues and eigenvectors for $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

Solution: 1. Characteristic polynomial:

$$\det(A - \lambda I) = \lambda^2 - 7\lambda + 10 = 0 \implies \lambda_1 = 5, \quad \lambda_2 = 2.$$

2. Eigenvectors:

- For $\lambda_1 = 5$: $(A - 5I)v = 0 \implies -v_1 + v_2 = 0 \implies v^{(1)} = (1, 1)^T$.
- For $\lambda_2 = 2$: $(A - 2I)v = 0 \implies 2v_1 + v_2 = 0 \implies v^{(2)} = (1, -2)^T$.

Answer: Eigenvalues: $\lambda_1 = 5, \lambda_2 = 2$; Eigenvectors: $v^{(1)} = (1, 1)^T, v^{(2)} = (1, -2)^T$.

12 Section 4. Bonus Problems and Theory (Assignment 3)

Bonus Problem 3.1

Problem Statement: Prove that if λ is an eigenvalue of an invertible matrix A , then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .

Proof: Given $Av = \lambda v$ ($v \neq 0$). Since A is invertible, $\lambda \neq 0$. Multiply by A^{-1} :

$$A^{-1}(Av) = A^{-1}(\lambda v) \implies v = \lambda A^{-1}v \implies A^{-1}v = \frac{1}{\lambda}v. \quad \square$$

—

Theoretical Questions

a) What is the dimension of a linear space?

Answer. The dimension of a space V ($\dim V$) is the maximum number of linearly independent vectors in V , or equivalently, the number of vectors in any basis of V . $\square$

b) How does matrix rank relate to row/column independence?

Answer. The rank of a matrix equals the maximum number of linearly independent rows (which always equals the maximum number of linearly independent columns). $\square$

Part IV

Assignment 4. Linear Operators and Quadratic Forms

13 Section 1. Linear Operators

Problem 1.1

Problem Statement: Find the matrix of operator A rotating vectors in $\mathbb{R}^2$ counterclockwise by $\phi = \frac{\pi}{3}$.

Solution:

$$M_A = \begin{pmatrix} \cos(\pi/3) & -\sin(\pi/3) \\ \sin(\pi/3) & \cos(\pi/3) \end{pmatrix} = \begin{pmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix}.$$

Answer: $M_A = \begin{pmatrix} 0.5 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 0.5 \end{pmatrix}.$

—

Problem 1.2

Problem Statement: Operator A maps $e_1 \rightarrow (2, 5)^T$ and $e_2 \rightarrow (-1, 3)^T$. Find image of $x = (3, -2)^T$.

Solution:

$$A(x) = \begin{pmatrix} 2 & -1 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 6 + 2 \\ 15 - 6 \end{pmatrix} = \begin{pmatrix} 8 \\ 9 \end{pmatrix}.$$

Answer: $A(x) = (8, 9)^T.$

14 Section 2. Quadratic Forms

Problem 2.1

Problem Statement: Construct the matrix for $Q(x_1, x_2, x_3) = 3x_1^2 + 5x_2^2 - 2x_3^2 + 4x_1x_2 - 6x_1x_3 + 2x_2x_3$.

Solution: Diagonal entries are x_i^2 coefficients; off-diagonals are half of x_ix_j coefficients:

$$A = \begin{pmatrix} 3 & 2 & -3 \\ 2 & 5 & 1 \\ -3 & 1 & -2 \end{pmatrix}.$$

—

Problem 2.2

Problem Statement: Test $Q(x_1, x_2) = 2x_1^2 + 6x_1x_2 + 5x_2^2$ for definiteness using Sylvester's criterion.

Solution: $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$. Principal minors: $\Delta_1 = 2 > 0$, $\Delta_2 = 10 - 9 = 1 > 0$.

Answer: Positive definite.

15 Section 3. Gram-Schmidt Orthogonalization

Problem 3.1

Problem Statement: Orthogonalize $f_1 = (1, 1, 0)$ and $f_2 = (1, 0, 1)$.

Solution: 1. $u_1 = f_1 = (1, 1, 0)$. 2. $u_2 = f_2 - \frac{\langle f_2, u_1 \rangle}{\|u_1\|^2} u_1 = (1, 0, 1) - \frac{1}{2}(1, 1, 0) = (0.5, -0.5, 1)$.

Answer: $u_1 = (1, 1, 0)$, $u_2 = (0.5, -0.5, 1)$.

16 Section 4. Bonus Problems and Theory (Assignment 4)

Bonus Problem 4.1

Problem Statement: Prove that eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal.

Proof: Let $A = A^T$, $Av_1 = \lambda_1 v_1$, $Av_2 = \lambda_2 v_2$ with $\lambda_1 \neq \lambda_2$.

$$\lambda_1 \langle v_1, v_2 \rangle = \langle Av_1, v_2 \rangle = v_1^T A^T v_2 = v_1^T A v_2 = \langle v_1, A v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle.$$

$$(\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0 \xrightarrow{\lambda_1 \neq \lambda_2} \langle v_1, v_2 \rangle = 0. \quad \square$$

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Theoretical Questions

a) State Sylvester's criterion for a negative definite quadratic form.

Answer. A form is negative definite iff its leading principal minors alternate signs starting with negative: $(-1)^k \Delta_k > 0$ for all $k = 1, \dots, n$ ($\Delta_1 < 0, \Delta_2 > 0, \Delta_3 < 0, \dots$). $\square$

b) What is a self-adjoint (symmetric) operator?

Answer. An operator A on an inner product space is self-adjoint if $\langle Ax, y \rangle = \langle x, Ay \rangle$ for all x, y . Its matrix in an orthonormal basis satisfies $A = A^T$. $\square$

Part V

Assignment 5. Singular Value Decomposition (SVD) and Tensors

17 Section 1. Singular Value Decomposition (SVD)

Problem 1.1

Problem Statement: Find singular values for $A = \begin{pmatrix} 3 & 0 \\ 0 & -4 \end{pmatrix}$.

Solution:

$$A^T A = \begin{pmatrix} 9 & 0 \\ 0 & 16 \end{pmatrix} \implies \lambda_1 = 16, \lambda_2 = 9 \implies \sigma_1 = \sqrt{16} = 4, \sigma_2 = \sqrt{9} = 3.$$

Answer: $\sigma_1 = 4, \sigma_2 = 3$.

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Problem 1.2

Problem Statement: Construct SVD $A = U\Sigma V^T$ for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Solution: $A^T A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Eigenvalues: $\lambda = \frac{3 \pm \sqrt{5}}{2}$. Singular values: $\sigma_1 = \frac{1 + \sqrt{5}}{2} \approx 1.618$, $\sigma_2 = \frac{\sqrt{5} - 1}{2} \approx 0.618$.

$$\Sigma = \begin{pmatrix} \frac{1 + \sqrt{5}}{2} & 0 \\ 0 & \frac{\sqrt{5} - 1}{2} \end{pmatrix}.$$

Vectors in V are normalized eigenvectors of $A^T A$, and columns of U are computed as $u_i = \frac{1}{\sigma_i} A v_i$.

18 Section 2. Pseudoinverse Matrices

Problem 2.1

Problem Statement: Find Moore-Penrose pseudoinverse A^+ for $A = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

Solution: Full column rank formula $A^+ = (A^T A)^{-1} A^T$:

$$A^T A = 2^2 + 1^2 = 5 \implies A^+ = \frac{1}{5} (2 \ 1) = (0.4 \ 0.2).$$

Answer: $A^+ = (0.4 \ 0.2)$.

19 Section 3. Tensors and Contraction

Problem 3.1

Problem Statement: Given $T \in \mathbb{R}^{2 \times 2 \times 2}$ with $T_{ijk} = i + j + k$, compute contraction along indices 1 and 2: $S_k = \sum_{i=1}^2 T_{iik}$.

Solution:

- For $k = 1$: $S_1 = T_{111} + T_{221} = (1 + 1 + 1) + (2 + 2 + 1) = 3 + 5 = 8$.
- For $k = 2$: $S_2 = T_{112} + T_{222} = (1 + 1 + 2) + (2 + 2 + 2) = 4 + 6 = 10$.

Answer: $S = (8, 10)^T$.

20 Section 4. Bonus Problems and Theory (Assignment 5)

Bonus Problem 5.1

Problem Statement: Prove that $(A^+)^+ = A$ for any matrix A .

Proof: The Moore-Penrose pseudoinverse $X = A^+$ uniquely satisfies: 1) $AXA = A$, 2) $XAX = X$, 3) $(AX)^T = AX$, 4) $(XA)^T = XA$. Let $Y = X^+ = (A^+)^+$. Y must satisfy identical conditions with respect to X : 1) $XYX = X$, 2) $YXY = Y$, 3) $(XY)^T = XY$, 4) $(YX)^T = YX$. Substituting $Y = A$ reduces these to the original four properties of X . By uniqueness, $(A^+)^+ = A$. $\square$

Theoretical Questions

a) What is the geometric interpretation of SVD?

Answer. Geometric action of A decomposes into: rotation/reflection (V^T), scaling along coordinate axes by σ_i (Σ), and a second rotation/reflection (U). It maps a unit sphere into an ellipsoid with semi-axes of length σ_i . $\square$

b) How does tensor rank differ from matrix rank?

Answer. Matrix rank equals the maximum number of independent rows/columns and is computable in polynomial time. CP-rank of a tensor order ≥ 3 is the minimal number of rank-1 tensors summing to it; computing it is NP-hard, and it can exceed any single spatial dimension. $\square$